



Events in Syntax: Modification and Predication

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Abstract

What are the motivations for including event variables in our representations of natural language semantics? The article focuses narrowly on the issues that provided the basis for the original discussion in Davidson (1967), and seeks to explore its consequences for the advances in syntax and the interface that have taken place since then. The existence of an event variable in semantic representations is argued to be as necessary as it ever was, but a more articulated version of the traditional view is proposed to account for the full range of phenomena such as scope of adverbials, aspectual modification, and aktionsart. A range of important architectural issues arise concerning the relationships between syntax, semantics, and the lexicon once the Davidsonian position is taken seriously, an area that has proved fruitful for investigation and debate. In Section 3, the article gives an example of one kind of answer to the architectural question by proposing a radically decomposed syntax that is tightly correlated with the semantics of event structure, a theoretical position that could be called ‘post-Davidsonianism’.

1 Introduction

Davidson (1967) first argued persuasively for admitting events as a crucial logical element in our understanding of natural language predications. These claims ushered in a productive and exciting period in the field, which saw event variables being exploited in various ways in semantic theories and in theories of the syntax-semantics interface (Higginbotham 1985; Kratzer 1989; Parsons 1990; Pustejovsky 1991; Kratzer 1996; Maienborn 2005; *inter alia*). The literature since 1967 that has crucially made use of events in understanding the semantics of natural language is huge and cannot be done justice in an article of this length. Events, once admitted into our ontology of entities, can be treated on a par with objects and can be manipulated as such in the semantic representations of various complex phenomena in natural language. Events have been central in the semantic understanding of pluractionality and distributivity on a par with plurality in the nominal domain (Schein 1993; Lasnik 1995; Landman 2000). Links between nominal and temporal reference and the construction of telicity have been discussed in important work by Verkuyl (1972, 1993), Filip (1999), and Krifka (1989, 1992), the latter exploring a mereological

approach based on the lattice theoretic framework of Link (1983). Events have also been exploited as discourse entities in theories of discourse representation as in Kamp and Reyle (1993). The important work done in these areas has shown, even more than conceptual argumentation could, the value of the event perspective on semantics, but a detailed examination of all the advances in our understanding made by this research is beyond the scope of any one article. This article has a somewhat narrower purpose: to revisit some of the phenomena that provided the initial motivations for admitting the event variable and to reconsider the consequences in the light of current theories of the syntax–semantics interface. In particular, it will examine the event variable qua ‘argument variable’ and assess its relationship to nominal referential variables, to event classification and modification.

In the first part of this article, I will briefly reprise the reasons initially proposed for admitting events as elements of the referential ontology in the expression of the semantics of natural language, a basic position that I do not take now to be seriously in doubt. On the other hand, many aspects of the formal theory of syntax and semantics have changed in the 40 years since the publication of Davidson’s seminal article, and event variables have been deployed in a wide variety of different ways. In the second part of this article, therefore, I will take the opportunity to critically discuss some of the issues that arise from this more recent research, and propose my own view on where we should be in the more complex ‘post-Davidsonian’ era.

The substance of the Davidsonian position is that while nominals are inherently object or matter descriptors, verbal predicates denote a different kind of individual that we might call an ‘event’ or ‘state of affairs’. Davidson argued that an event variable should be present in the semantic representation of verbal meaning, as one of its essential ‘arguments’. The kinds of representations Davidson argued for look as in (1) below.

- (1) Jones buttered the toast.
 $\exists e[\text{buttering}(e, \text{Jones, the toast})]$

In other words, there is an event that is a ‘buttering’, that has ‘Jones’ as its first argument, and ‘the toast’ as its second argument.

Note that the event position introduced by a predicate is functionally and logically distinct from the argument positions traditionally thought to be occupied by selected (usually DP) objects. Whereas the event position is taken to be the variable that the verb somehow describes by its lexical content, the DP argument positions are constitutive participants in that event and are selected by the verb. The verb cannot ‘select’ an event variable – it simply denotes a property of events.

The existence of this event variable in the logical semantics is justified, according to Davidson, on a number of grounds. First, it appears that human language can actually refer to events directly by means of pronominal reference (2), and they can even be quantified over (3).

- (2) Jones buttered the toast. It was over very quickly.
 (3) In every burning, oxygen is consumed. (from Parsons 1990: 18–19)

More central to the Davidson's original argument is the problem of valency of event predications. Even apart from nominal arguments, verbs can be modified by an unpredictable number of linguistic phrases, all of which have to somehow be integrated into the description of the event. The variable polyadicity for a verb like 'butter' in the examples below (from Davidson 1967) is not tenable, because it fails to generate the right entailment relations among all the examples in (4).

- (4) a. Jones buttered the toast.
 b. Jones buttered the toast slowly.
 c. Jones buttered the toast slowly in the bathroom.
 d. Jones buttered the toast slowly in the bathroom with a knife.
 e. Jones buttered the toast slowly in the bathroom with a knife at midnight.

Thus, in the examples above (4e) entails (4a), (4b), (4c), and (4d); (4d) entails (4a), (4b), and (4c); (4c) entails (4a) and (4b); (4b) entails (4a). Alternatively, if the logical representation of (4e) above employs an event variable common to all the modifiers, then the different modifiers can be linked by simple conjunction and then the entailments follow as a natural logical consequence (5).

- (5) Jones buttered the toast at midnight with a knife.
 $\exists e[\text{buttering}(e, \text{Jones, the toast}) \ \& \ \text{Slow}(e) \ \& \ \text{in}(e, \text{the bathroom}) \ \& \ \text{With}(e, \text{a knife}) \ \& \ \text{At}(e, \text{midnight})]$

The formula in (5) states that there exists an event that is a 'buttering' of 'the toast' by 'Jones', that it was 'slow', that the event was 'in the bathroom', that it was 'with a knife', and that it was 'at midnight'. Thus, Davidson's account allows the adjuncts to enter the semantic representation in a truth functionally compositional way, to create a complex event description.

The traditional Davidsonian representation above, however, tends to obscure the logical and linguistic difference between the event variable, the entity that the verbal description denotes, and the arguments denoting the selected participants in that event. Moreover, if there are indeed generalizations over participant role types across predicates and across languages, then the traditional Davidsonian representation does not have straightforward way of classifying the semantic participanthood for each of the different selected arguments. The neo-Davidsonian representation, on the other hand (see Parsons 1990), expresses each participant relation via its own thematic predicate, and severs those arguments from the verb description *per se*.

- (6) Jones buttered the toast.
 $\exists e[\text{buttering}(e) \ \& \ \text{Agent}(e, \text{Jones}) \ \& \ \text{Theme}(e, \text{the toast})]$

Here, we say that there exists an event, where the event is a 'buttering', the agent of the event is 'Jones', and the theme of the event is 'the toast'.

If the neo-Davidsonian agenda is to be pursued, it must therefore embed an adequate theory of thematic role labels. For many purposes, the exact nature of the labels and their definitions can be ignored. Unfortunately, it has been a difficult problem to solve, as reliable entailments over the proposed content of traditional labels have proved illusive (see Dowty 1989 for an important discussion). Many authors now favor an extremely reduced, more event oriented set of roles for structurally licensed positions in the clause either in addition to, or completely supplanting traditional theta role labels (Grimshaw 1990; Jackendoff 1990; Baker 1997). Baker (1997), for example, proposes a classification that includes just agent, theme, and goal/path, each with its own distinguished syntactic position. I will return to the issue of thematic participants in the context of event decomposition in later sections, where the abstract participant types I will argue for correspond fairly closely to this more reduced inventory of role types.

When it comes to adverbial modification, the proposal is that adverbials directly introduce predicates over events as well, and that syntactic modification corresponds to thematic identification of the two event variables involved (cf. Higginbotham 1985). This is assumed to give rise to a conjunction of event predications, as in (7).

- (7) Jones buttered the toast slowly.
 $\exists e[\text{buttering}(e) \ \& \ \text{Agent}(e, \text{Jones}) \ \& \ \text{Theme}(e, \text{the toast}) \ \& \ \text{Slow}(e)]$

So, now in addition to the facts expressed by the formula in (6), we also know that the event was 'slow'.

Seen in this way, the event position is not an 'argument' of the verb in the sense of required participant, but more analogous to the individual variable introduced by a noun: the noun expresses a predicate over things. (The formula below simply states that 'book' is a predicate that is true precisely of those things that are books, and false otherwise.)

- (8) 'book' : $\lambda x[\text{book}(x)]$

This is actually the substance of the Davidsonian position: while nominals are inherently object or matter descriptors, verbs denote a different kind of individual that we might call 'states of affairs'.

Related to this issue, it is important to point out the intermediate position of Angelika Kratzer with respect to neo-Davidsonianism, which she describes as semi-Davidsonian. In her system, only the 'external' argument is actually severed from the verb (Kratzer 1996), while the internal one remains a part of the verbs own argument structure. For our 'toast buttering' example, the representation would look as in (9) below.

- (9) Jones buttered the toast.
 $\exists e[\text{buttering}(e, \text{the toast}) \ \& \ \text{Agent}(e, \text{Jones})]$

Another important point of disagreement is the extent to which all verbal predicates actually have an eventuality argument in their lexical representation. Some researchers have argued that the event position is present for all predications involving situations that have some temporal sensitivity, even those that may be stative, or even nominal (e.g. Higginbotham 2000). Others wish to restrict it to certain types of verbal predications, crucially excluding some states (e.g. Kratzer 1989; Maienborn 2005).

In an extremely influential article, circulated in the late 1980s, but originally only published in 1995, Kratzer proposed that some types of stative predications ('stage-level' states in the sense of Carlson 1977), carried an eventuality variable in their logical semantics, while others ('individual-level' predications) did not (Kratzer 1995). The evidence for this position was the linguistic behavior of two different types of verb phrase with respect to certain adverbial modifications and modification by tense. In particular, 'stage-level' predications showed evidence of a spatio-temporal variable that could be bound, predicated over and quantified over, while 'individual-level' predications did not. Consider the following examples, which are representative of the set of contrasts Kratzer describes. The flexible word order in German gives rise to ambiguity in the case of the stage level sentence in (10) as the locative phrase can either describe the adjacent noun phrase directly, or modify the event as a whole, saying something about the spatio-temporal location of the event. In (11), by hypothesis, there is no such spatio-temporal variable available to be modified and the sentence has only NP modifying reading.

- (10) ... weil fast alle Flüchtlinge in dieser Stadt ungekommen sind.
 ... since almost all refugees in this city perished are
 (a) 'since almost all of the refugees in this city perished'
 (b) 'since almost all the refugees perished in this city.'
- (11) ... weil fast alle Schwäne in Australien schwarz sind.
 ... since almost all swans in Australia black are
 (a) 'since almost all swans in Australia are black.'
 (Kratzer 1995: 127).

In Ramchand (1996) and Higginbotham and Ramchand (1997), we tried to argue that although the linguistic distinctions were important, they did not point to a lexical, but rather a constructional distinction, and that there was no reason to suppose that any verbal predicate lacked an eventuality variable in principle. de Swart (1996) similarly argues that a stative variable might independently be incapable of modification by adverbs such as 'whenever', without forcing the conclusion that the situational variable does not actually exist for these forms. Maienborn (2005) argues that some states, but not others, should be endowed with a situational variable in their representation, and the argument there is also based on modifications. At the same time, Maienborn still has to invoke a new

kind of variable (the 'Kimian' state variable) to account for certain predication phenomena related to her 'non-Davidsonian' states. While the issue is certainly not settled, I will continue to assume that all relational predicates require some variable for 'states of affairs' that is necessary for building up a complex description in the first place.

In the 1980s and early 1990s, the use of event variables in syntactico-semantic representations was couched within a framework that had a well-developed theory of the lexicon as the locus of argument structure, subsequent work has taken a more constructional view of all these argument structure issues (Borer 2005; Ramchand 2007). The difference between stage level and individual level predications is intimately tied up with this debate. I will return to the issue of how Davidsonianism pans out in a more constructional approach to predicate argument relations in Section 3.

Another important area where debate over the existence of eventuality variables has been controversial (and fruitful), has been in the domain of nominalizations. In an important early discussion, Grimshaw (1990) showed that the argument structure of derived nominals was related in a systematic way to properties of their verbal counterparts, showing the centrality of event properties of predicates even in the category change from V to N. In that work, nominalizations based on stative verbs could be shown to never possess obligatory arguments, unlike dynamic event denoting nominals. Szabolcsi (1994) argued that this could be explained by saying that statives possessed no eventuality variable, if arguments are introduced by the event itself. Once again, the resolution of these issues is bound up with architectural questions concerning the role of syntax in the realm of argument structure (and category changing) morphology, and in the proper treatment of the semantic and predication differences between the categories of N and V. A detailed discussion of these issues is beyond the scope of this article, but see Borer (2005) for a recent treatment in a constructional framework. In what follows, I will continue to assume the existence of an eventuality variable in predications involving the syntactic category Verb (V), regardless of aspectual differences. In fact, as we will see in Section 3, complex aktionsart structure will be argued to be decomposable into processual and stative components, a move that would not be articulable in a theory that possessed no purely stative eventuality variable types.

2 *Events and Identity*

The Davidsonian agenda opens up the philosophical problem of articulating 'identity criteria' for events, a notoriously difficult task (Quine 1960; Kim 1966, 1976; Hacker 1982; Parsons 1990; Asher 1998; Carlson 1998). Simply identifying events by their occupation in space and time is not sufficient because 'John pushing a cart' and 'John's muscles expanding and contracting in a particular manner over exactly the same time span circumscribed by the pushing' do not count as the 'same' event, as far as natural

language is concerned. It is less easy to decide whether Brutus' stabbing of Caesar is the 'same event' as Brutus' brutal stabbing of Caesar (see Higginbotham 2005 for an argument that they are). Such indeterminacies might be seen by some as undermining the general plausibility of event variables as part of our logical ontology. Maienborn (2005) claims that 'crucial for our notion of eventualities . . . is their inherently relational character', although she still attempts to ground them, or justify them in terms of objective real world criteria. In fact, I would argue that it is an entirely reasonable position to claim that eventuality variables, as linguistic entities, do not denote real objects embedded in space and time with inherent identity criteria. Rather, they are extremely abstract relational objects that are built up compositionally, defined only by means of their constitutive participants (see Schein 2002 for a discussion of the effects of changing the event description on a theory of thematic role types). The linguistic approach to the existence of events has been largely to ignore the metaphysical issues involved. Instead, as the evidence from language is that we refer to them, quantify over them, reason with them, and use them to build up propositional statements about the world, their existence is justified within the representational systems we employ to describe the semantics of natural language.

This difference between the categories N and V is potentially significant for theories of grammar, if it is true that objects indeed have more stable identity criteria by some objective psychological measures than events. It has recently been argued that the deep difference between the two syntactic categories N and V is precisely that verbs do not have independent identity criteria (cf. Baker 2003), but must build up states of affairs from their constitutive participants. One upshot of this is that if verbs denote event descriptions, they are dependent on the expression of their constitutive participants in a way that non-derived nominals are not. The syntactic consequence of this is that verbs require arguments while (underived) nominals do not.

2.1 CLASSIFICATION OF EVENTS

Taking it for granted that events in the world do not necessarily correspond to a single objective linguistic description, it can be noticed that languages distinguish between different subtypes of event conceptually, along dimensions of internal complexity with regard to change and time. It is important to emphasize that these subtypes of event are distinguished in the literature because of the ways in which they behave linguistically and morphologically in various languages. Perhaps one of the most influential early articles on work on aktionsart is Vendler's (1967), in which he argued for the difference between dynamic events on the one hand, and nondynamic events (states) on the other.¹ Furthermore, within the dynamic events, we countenance a distinction between simple processes (activities)

that also include a *telos* (accomplishments) and perhaps also punctual transitions (achievements), all of which exhibit different behaviors with respect to linguistic diagnostics (see Dowty 1979). However, it is crucial that all of these different types of events have not so far been considered distinct ontological categories. It is now also clear that certain event types seem to be complex in that they contain more than one subevent. Accomplishments, for example, are now commonly assumed to consist of both a process subevent and a result subevent (Parsons 1990; Pustejovsky 1991; Higginbotham 2001). Once again, the accomplishment verb type is not a distinct ontological category, but a complex event form constructed from ontological entities we already acknowledge to exist.

2.2 HOW MANY EVENT VARIABLES?

Outside of the existence of thematic participants, which form the necessary core of the event description, we see that a verbal event can also be modified by higher aspectual operators, and, moreover, that adverbials can occur at various hierarchical heights in the syntactic structure. Following the logic of the arguments based on modification, different levels of eventuality variables seem to be required here. For example, consider the addition of the progressive auxiliary in English (12b), and the habitual past tense (12c). These create descriptions that have different internal temporal properties from that expressed by the verb alone. Thus, the complex event of habitually crossing a street has different constitutive properties and different co-occurrence restrictions with modifiers than the simple noniterated accomplishment version in (12a).

- (12) a. John crossed the street (in two minutes).
 b. John was crossing the street (*in five seconds/for only five seconds, before he changed his mind).
 c. John crossed that street for years on his way to work.

While it might be argued that the special properties of the progressive, or habitual operators, are what is at stake here, it is equally possible that a more radical approach to the eventuality variable is called for, with the various aspectual operators introducing their own derived eventuality descriptions based on the event predicates they combine with [this is essentially the approach taken by Parsons (1990) in his analysis of the perfect and the progressive in English]. Modern research at the syntax-semantics interface and on the interpretation of an ever increasing number of functional projections makes it more difficult to think of the event as a monolithic entity that is introduced once and for all by a verb. The structure of the clause is such that the event variable is constantly being modified and manipulated, and that various functional positions in the clause are semantically responsible for introducing those changes (cf. de Swart 1998; Kratzer 1996; Ernst 2002).

Maienborn (2005), following on from criticisms in Bennett (1988), raises concerns about what she calls the 'latent infinite regress of eventualities'. Consider the neo-Davidsonian representation of the modified buttering event in (13) below.

- (13) Jones buttered the toast at midnight with a knife.
 $\exists e[\text{buttering}(e) \ \& \ \text{Agent}(e, \text{Jones}) \ \& \ \text{Theme}(e, \text{the toast}) \ \& \ \text{At}(e, \text{midnight}) \ \& \ \text{With}(e, a \text{ knife})]$

The classical view presented above takes the eventuality variable *e* that is modified by 'at midnight' to be the very same one that is modified by the instrument phrase 'with a knife'. But, as Maienborn correctly asks: 'But what prevents us from taking the fact that *e* has a certain property to introduce another eventuality? . . . creating an infinite regress of eventualities?' (2005: 311). This is a legitimate point, but in the approach advocated here, not necessarily a problem. I have already suggested that at different levels of syntactic structure, updated eventuality variables do need to be constructed. While it may be true that there is no inherent limit to the number of different adverbs we might add to a simple verbal predicate *in principle*, there is no danger of actual infinite regress as in practice the topmost eventuality variable will be existentially bound creating a well-formed proposition. Updating eventuality variables is precisely what we need to be able to do in a compositional syntax that contains far more structure than was originally envisaged in the 1960s. Moreover, this will potentially allow us to capture certain scopal effects of modifiers that would not be easy to state under the classical neo-Davidsonian system. While the adverbs in (13) above operate on fairly independent semantic dimensions, the examples in (14) may be more telling.

- (14) a. Jones buttered the toast quickly with the tiny knife.
 b. Jones buttered the toast with the tiny knife quickly.

In (14a), it was the buttering of the toast that was quick (for a toast-buttering), and, moreover, it was done with a tiny knife – how impressive of Jones to achieve the requisite speed with such a small implement! On the other hand (14b) means that the quickness of Jones action is relativized to toast-butterings with tiny knives, and is perfectly consistent with that action taking longer than it would have if he had had a normal sized butter spreader. There might be ways of capturing the different readings here, by directly encoding a position in the argument structure of the modifier 'quickly' that denotes the property with respect to which the modifier is relativized: the (14b) reading would involve relativizing 'quickly' to the property of being 'a toast-buttering with a tiny knife'; on the (14a) reading, 'quickly' is only relativized to 'toast-buttering'. However, this would require an explicit argument position for such a property [as indeed proposed by Higginbotham (1985) as autonymous theta marking],

and a theory concerning how this property is chosen based on the syntactic structure. On the other hand, constructing derived eventuality variables at different points of the syntactic structure gets these facts to fall out directly from compositionality. Examples like this, and many other cases where the actual positions of the adverbials are forced in a particular order, make it plausible that examples such as in (13) and (14) should be represented using at least two, but related, eventuality variables, as shown in (15).²

- (15) Jones butter the toast quickly with a tiny knife.
 $\lambda e''\exists e',e$ [buttering(e) & Agent(e, Jones) & Theme(e, the toast) & Constitutive-Event(e',e) & quickly(e') & Constitutive-Event(e'',e') & With(e'',a tiny knife)]

The formula above says that there is an event of 'buttering' that has Jones as its agent, and the 'toast' as its patient, and that this is a crucial subcomponent of a derived event e' , which exists and which is defined by additional property that it is also a quick event. This derived quick event e' is in turn a constitutive part of a third derived event e'' , which is defined by the fact that it is an instance of e' that takes place with a tiny knife. The resulting predication is now a description of the complex event e'' , which is dependent on the other embedded event descriptions.

Under this view, successive constituents in the verbal extended projection still each denote predicates over events, but not the *same* event variable each time. The event variable is modified and updated as one builds up the meaning of the syntactic structure from bottom to top, as is plausibly the case if we are to admit that events are relationally defined. Once all participants and event modifiers have been incorporated into the description of the event, the syntactic projection so constructed will denote a large complex event. It is this event description that will be input to all further semantic manipulations, such as tense, that will eventually construct a full proposition. This view is thus more articulated than the original position in the literature, but its central assertion is still that the event variable is the linguistically real and active heart of the clause.

On the issue of which syntactic elements require a Davidsonian variable in their representation, I take this to be an empirical point, and may well be found to be correlated with deep differences in syntactic category. Diagnosing whether a particular natural language form is event denoting or object denoting will depend on whether they are best interpreted as inherently relational states of affairs linguistically, or as autonomously identifiable objects.

3 *The Relation Between Events and the Syntax*

From quite early on, connections between event semantics and syntactic representation have been proposed. One important issue concerns how the eventuality variable is bound (on analogy with object variables in logical representations). The well-established view in the literature (cf.

Higginbotham 1985) is that the place in the syntax where tense relations are expressed (the tense node, or inflectional node in earlier theories) is the level at which the eventuality variable is bound by existential closure (what Higginbotham 1985 calls 'theta binding'). Under the more articulated view, I have discussed in the previous section, the event variable is continually existentially bound and updated with a new one. But it still remains true that at some point, embedding under the higher structures of the clause allows the maximal complex event to be existentially closed off and located in time in order to construct a proposition.

It seems clear that an event description is intimately connected to temporal reference, even though it is not defined by its temporal trace. Thus, in the semantics, temporal trace functions have been assumed that map an event to its run time (e.g. Krifka 1989, 1992); in the syntax, the verbal projection is syntactically selected by T (Baker 2003), actual verbs showing morphological marking for tense in many languages.³

I would like to defend an emergent view in the field that is moving away from the autonomous lexical view of early Davidsonianism and towards a more generative-constructivist one, taking seriously recent advances in our syntactic understanding of the functional hierarchies of the clause. This is the view that I have been calling post-Davidsonianism. Once compositional subtleties are acknowledged, the connections between the syntax and the semantics of events become even more interesting. In the next two subsections, I focus on two aspects of the internal complexity of the linguistic event, arguing that they call for an articulation of verbal syntax.

3.1 RESULTATIVITY

To take an example, the simple manner verb 'hammer' is consistent with an atelic interpretation, or more accurately, with an interpretation whereby no set result state is implied (16a), whereas the addition of a resultative adjectival predicate creates a more complex accomplishment in the syntax (16b).

- (16) a. John hammered the metal.
b. John hammered the metal flat.

The logic of eventuality decomposition demands that the event position of the former sentence is topologically and linguistically different from the complex event position of the latter, having different constitutive properties.

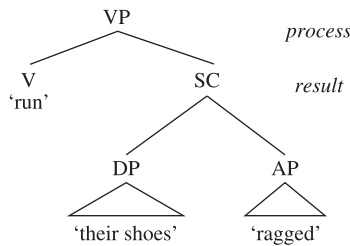
The internal complexity of (some) events is not a novel position, but it is one that has usually been expressed in lexicalist terms (Jackendoff 1990; Pustejovsky 1991; Levin and Rappaport Hovav 1995; Rappaport Hovav and Levin 1998). Rappaport Hovav and Levin (1998) call the process by which an activity verb gains a result state in its event template 'template augmentation', because of the addition of a resultative subevent creating an accomplishment from a pure activity. However, as resultative formation manipulates syntactic primes (separate lexical words), it is also possible

to see resultativity as something that is essentially constructed syntactically. This strategy takes seriously the independent predication properties of the secondary predicate (here an AP), which are especially striking in the case of ‘unselected’ objects of verbs whose events have been complexified by the addition of extra predication structure. The example in (17) shows that the addition of the secondary predicate ‘ragged’ gives rise to a previously unselected internal argument ‘the shoes’, hence the term unselected object.

- (17) a. They ran their shoes ragged.
b. ★They ran their shoes.

The idea here is that the secondary predicate introduces its own small clause with predication subject. Thus, ‘the shoes’ are simultaneously the subject/holder of the property of being ‘ragged’ as well as the direct object of the verb, receiving accusative case. To get the composition to work out right, we also need to ensure that the small clause so formed is interpreted as the result of the activity expressed by the main verb (see Hoekstra 1988, and Hoekstra and Mulder 1990 for discussion).

(18)



The small clause projection is a stative eventuality description that composes directly with the activity eventuality description contributed by the main verb. Thus, the syntax can mimic the effects of ‘template augmentation’, which was assumed by Rappaport Hovav and Levin (1998) to apply in the lexicon, provided certain assumptions about the mapping between syntax and semantics are made. I will return to this issue below.

Theories that assume some form of template augmentation in the lexicon run into paradoxes if argument structure is supposed to be an encapsulated level of representation that is presyntactic. Expanding the combinatorial possibilities in the lexicon to include the composition of syntactically independent formatives such as ‘run’ and ‘ragged’ runs the risk of duplicating the principles of syntax, and obscures generalizations about complex predication that are related to those syntactic facts. For example, Ramchand (2007) argues that analytic V–V (resultative) complex predicates in Bengali and Hindi/Urdu need to be formed according to syntactic structuring principles and straightforwardly reflect the ‘head final’ phrase structural properties of those languages.

3.2 CAUSATION

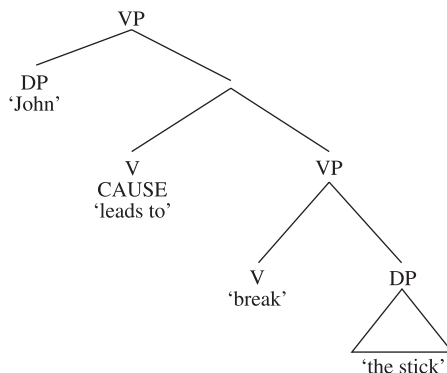
The other aspect of event structural complexity that has been prominently treated in the literature is that of causation. Many languages have verbs that alternate (with or without the addition of extra morphology) between causative and inchoative versions. A large class of verbs in English called ‘ergatives’ in Hale and Keyser (1987) conform to this pattern.

- (19) a. The stick broke.
b. John broke the stick.

Once again, it is very common to find lexical treatments of this alternation, treatments that essentially rely on purely argument structure manipulations instead of directly invoking subevental complexity (Levin and Rappaport Hovav 1995; Reinhart 2002; Chierchia 1998).⁴ The argument structure approach has also been syntacticized, however, exploiting the long appreciated differences between internal and external arguments (Marantz 1984; Williams 1982). In particular, the phenomenon of VP shells [initially proposed by Larson (1988) in his analysis of double object constructions] has been employed by various people in the guise of a little *v* functional head that is responsible for introducing the external argument (Hale and Keyser 1993; Kratzer 1996; Harley 1995; Pylkkänen 1999). In the case of Kratzer (1996), ‘severing the external argument from the verb’ is a move that is reflected both in the syntax she assumes and the semi-Davidsonian representations she gives, thus giving a pleasing mapping between syntactic and semantic representations.

The Hale and Keyser’s (1993) view is an interesting one to examine in this regard (even though it hedges somewhat on the distinction between lexicon and syntax), because it comes closer to a more eventive decompositional view of transitivity: the higher V head is labelled CAUSE, and the higher V is linked to the lower V by the ‘leads to’ relation. A version of the trees proposed by Hale and Keyser is given below in (20).

(20)



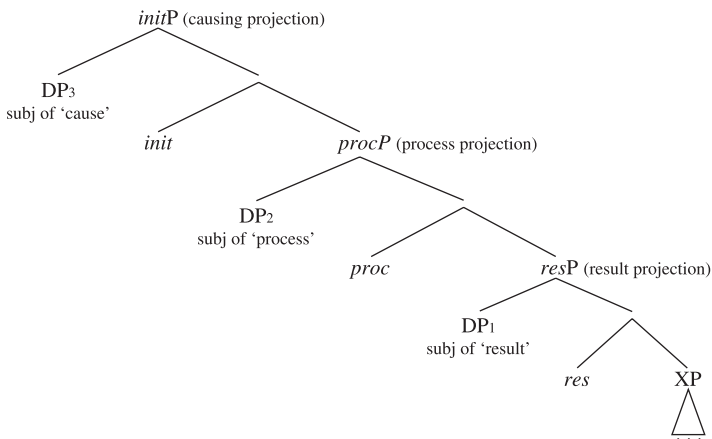
In the next subsection, I propose a syntactic decomposition that integrates both these traditions of decomposition into a single representation of event structure complexity that makes event structure primary and argument structure secondary to the way in which the syntax reflects meaning.

3.3 LINING UP EVENT STRUCTURE WITH SYNTAX

At this point in the literature, there are many proposals for articulated syntaxes of the verb phrase that syntacticize some or all of the elements that were traditionally considered to be the purview of lexical meaning (Ritter and Rosen 1998; Travis 2000; Borer 2005). These proposals are designed to account for argument licensing and facts about telicity primarily, and are an important part of the lexicon vs. syntax debate. The proposal made in Ramchand (2007) here is slightly different from those in that it takes event structure and predication complexity as its primary motivation, presented here in the context of the discussion about events as the most extreme claim about the tight relationship between syntax and the semantics of events. In a sense, it takes the Kratzerian position on the pleasing parallelism between syntax and semantic representations, but offers further elements of decomposition on both levels – dislocating *all* argument positions and relating them to their individual subevents.

This event structure syntax contains three important subevental components: a causing subevent, a process denoting subevent, and a subevent corresponding to result state. Each of these subevents is represented as its own projection, ordered in the hierarchical embedding relation as shown below in (21).

(21)



In (21) above, the label *init* (for initiation) has been chosen to represent the outer causal projection that is responsible for introducing the external argument; in many ways, it is similar to the external argument introducing ν as invoked in the recent literature (Hale and Keyser 1993;

Harley 1995; Kratzer 1996). The central projection that represents the dynamic process is called *procP* (for process phrase). The lowest projection has been given the label *res* for result. This decomposition stays away from more traditional categorial labels, such as V, to emphasize the fact that this system is actually a splitting up of what we normally think of as V, in the same spirit as Rizzi (1997)'s splitting up of the C node to show its fine structure, or the splitting of INFL into distinct heads 'Tense' and 'Agr' in Pollock (1989). All three of these projections are essentially verbal, and no individual piece actually corresponds to what traditional accounts would label V: the notion of verb is always a composite that involves some or all of these elements. Also, the tree above represents the maximal possible decomposition, and a dynamic verbal projection may exist without either the *init* or *res* elements.

Under this view, *procP* is the heart of the dynamic predicate, as it represents change through time, and it is present in every dynamic verb. In other words, a *procP* is present regardless of whether we are dealing with a process that is extended (i.e. consisting of an indefinite number of transitions) or the limiting case of representing only single minimal transition such as that found with 'achievement' verbs. The *initP* exists when the verb expresses a causational or initiational state that leads to the process. The *resP* only exists when there is a result state explicitly expressed by the lexical predicate; it does not correlate with semantic/aspectual boundedness in a general sense. Specifically, the telicity that arises because of the entailments based on DP structure and the verbal relation do not mean that *resP* exists, that is, *resP* only exists if the event structure itself is specified as expressing a result state. Conversely, the expression of result can be further modified by auxiliaries, PPs etc outside the core event building domain to create predications that are atelic, but this will not warrant the removal of *resP* in the syntactic representation.

The idea behind this more full-blown representation is that both causativity and resultativity require their own subevents within a complex macro event. Thus, while the English verb 'dry' can represent a pure process (22a), it can also be used to construct a predication that leads to a result (22b), a caused process (22c), or even both together (22d).

- (22) a. The cocoa beans dried on the racks (for three days).
- b. The cocoa beans dried bone dry (in just two days).
- c. The farmers dried the cocoa beans (for three days).
- d. The farmers dried the cocoa beans bone dry (in just two days).

Ramchand (2007) argues that the distribution of verbal lexical items can be captured by which syntactic category features they possess ([proc], in the case of 'dry') together with the availability of causational lexical items (null in English), and strategies for result licensing. A detailed discussion of the licensing relations between verbs and the syntax is beyond the scope of this short article, but see Ramchand (2007), Chapter 4, for a

description of the different verb classes of English. It is also argued there that other languages show more direct morphological evidence for these decompositions than English, whose verbs most commonly identify more than one structural position in the decomposition.

Once we decompose according to event predication structures, we can make participant semantics (thematic roles in traditional terminology) parasitic on event structure positions. This gives us a different, and more abstract, set of natural classes for participant roles [although they show important convergence with Baker's (1997) more impoverished classification, as described earlier]. Each projection represented here forms its own core predication structure with the specifier position being filled by the 'subject' or 'theme' of a particular (sub)event, and the complement position being filled by the phrase that provides the content of that event. The complement position itself of course is also complex and contains another minipredication, with its own specifier and complement. In this way, the participant relations are built up recursively from successively embedded event descriptions and 'subject' predications.

Thus, the semantic representation for a complex verbal predication like 'The farmers dried the cocoa beans bone dry' would look as in (23) below.

- (23) $\lambda e \exists e_1, e_2, e_3 [e = (e_1 \text{ LEADS-TO } (e_2 \text{ LEADS-TO } e_3)) \ \& \ [\text{initiate}(e_1) \ \& \ \text{process-dry}(e_2) \ \& \ \text{bone-dry}(e_3) \ \& \ \text{'subject'}(e_1, \text{the farmers}) \ \& \ \text{'subject'}(e_2, \text{the cocoa beans}) \ \& \ \text{'subject'}(e_3, \text{the cocoa beans})]$

The above formula is the description of a complex event e , which consists of three subevents e_1 , e_2 , and e_3 , such that e_2 leads to e_3 , and e_1 leads to the e_2 -to- e_3 combination. The subevents have the properties that e_1 is an initiatory event whose subject is 'the farmers', e_2 is a process of drying event whose subject is 'the cocoa beans', and e_3 is a resulting event of bone dryness whose subject is also 'the cocoa beans'.

The macro event e , is only composed at the top of the syntactic structure, syntactic embedding corresponds to the LEADS-TO relation⁵ and DPs in specifier position are the individual arguments introduced by each event description. An important aspect of this proposal is the claim that there is a general combinatorial semantics that interprets this syntactic structure in a regular and predictable way. Thus, the semantics of event structure and event participants is read directly off the structure, and not directly off information encoded by lexical items. For convenience, we can define labels for the predication 'subjects' of each individual subevent.

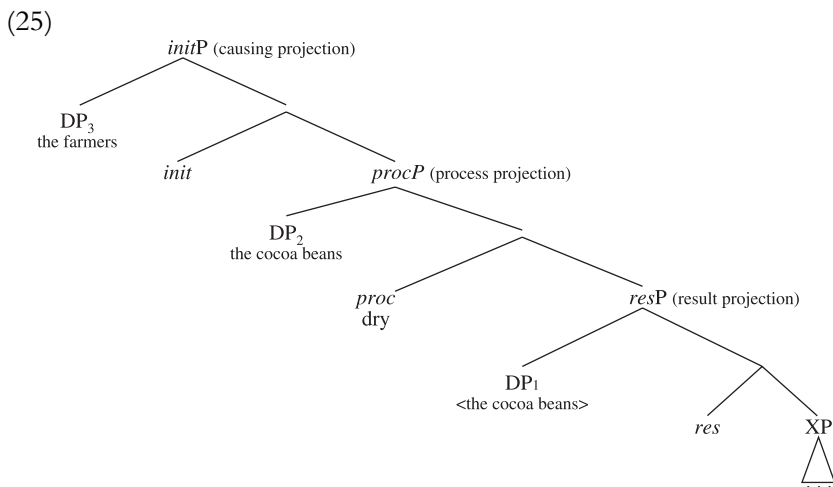
- *initP* introduces the causation event and licenses the external argument ('subject' of cause = INITIATOR)
- *procP* specifies the nature of the change or process and licenses the entity undergoing change or process ('subject' of process = UNDERGOER)

- *resP* gives the ‘telos’ or ‘result state’ of the event and licenses the entity that comes to hold the result state (‘subject’ of result = *RESULTEE*).

We can thus rewrite the semantic representation in (23) as (24).

- (24) $\lambda e \exists e_1, e_2, e_3 [e = (e_1 \text{ LEADS-TO } (e_2 \text{ LEADS-TO } e_3)) \ \& \ \text{initiate}(e_1) \ \& \ \text{process-dry}(e_2) \ \& \ \text{bone-dry}(e_3) \ \& \ \text{INITIATOR}(e, \text{the farmers}) \ \& \ \text{UNDERGOER}(e, \text{the cocoa beans}) \ \& \ \text{RESULTEE}(e, \text{the cocoa beans})]$

These labels each correspond to the DP filling a particular syntactic position. As can be seen from the representation in (24), a single DP can fill more than one predicational position under the decompositional view, giving rise to ‘composite’ thematic relations (‘the cocoa beans’ are simultaneously the UNDERGOERS of drying and the RESULTEE of the ‘bone-dry’ state). The syntactic representation corresponding to the semantic representation given in (24) is as in (25), with composite thematic relations derived by movement.



In the above, I have only illustrated the role types that emerge from predicational subject positions; in Ramchand (2007), it is also assumed that DPs can fill complement positions in this structure. In complement position, DPs codescribe the event predicate under conditions of homomorphism. In particular, the class of ‘incremental theme’ objects (cf. Krifka 1989, 1992) much discussed in the literature, as well as ‘path’ objects are argued to be found in this position. A detailed treatment of this, however, is beyond the scope of this article.⁶

4 Conclusion

In this article, I have moved from an initial justification of event variables in our ontology for semantic representations to discussing a wide variety

of issues that emerge once that position is taken seriously. Syntactic theory has changed a lot since 1967, but event variable remains at the heart of much of the most important work at the interface of syntax and semantics. I have only been able to touch on the various areas of interest here; they have included representational questions for adverbial modification, aspectual modification, causation, resultativity, and participant relations in verbal predication. Event semantics has made it possible to explore these, and other phenomena in natural language, where there seem to be pervasive and powerful generalizations.

In addition, the issue of events in linguistic representation interacts with long-standing architectural debates concerning the role of the lexicon in grammar, and the relation between syntactic and semantic representations. The view I have presented here places event semantics and thematic relations firmly within the nonlexical camp, and pursues a particularly tight mapping between syntactic and semantic representations. I have focused on this aspect of the problem because syntactic phrase structural representations have changed so radically in the last 40 years that it is not clear what a modern Davidsonianism would need to look like in a strong theory of the syntax-semantics map. I hope I have shown that the Davidsonian position generalizes in fairly obvious ways once more articulated syntactic and semantic representations are postulated. More importantly, what emerges from an examination of the literature is that the intuition behind the research programme inaugurated by Davidson (1967) is still fueling some of the most exciting and productive research at the syntax-semantics interface today.

Short Biography

Gillian Ramchand pursues a research programme concerning the relationship between syntactic and semantics representations of natural language. She has worked in areas of tense, aspect, predication and argument structure on languages as diverse as Bengali, Scottish Gaelic, and English. She has published articles in *Natural Language Semantics*, *Linguistic Inquiry* and *Lingua* as well as a number of edited volumes. She is the author of two books *Aspect and Predication* (Oxford University Press 1997) and *Verb Meaning and the Lexicon* (forthcoming, Cambridge University Press) where she argues for a syntactic implementation of an even structural view of verbal meaning and participant relations. Ramchand is currently Professor of Linguistics at the University of Tromsø, Norway, and a research member of the Center for Advanced Study in Theoretical Linguistics (CASTL) there. Before moving to Norway in 2003, Ramchand was lecturer in General Linguistics at the University of Oxford. She holds a PhD in Linguistics from Stanford University, and Bachelors degrees in Mathematics, and Philosophy, from the Massachusetts Institute of Technology.

Notes

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¹ As noted in Section 1, I will assume in what follows that states are an important *subtype* of eventuality variables, rather than a subcase where eventuality variables are not relevant.

² I assume that in certain cases, and within a certain local domain of modification, the straightforward conjunctive strategy from classic neo-Davidsonianism might be appropriate. At higher levels of modification, a 'new' more complex event is formed with internal structure, allowing scopal facts and selectional facts to be accounted for. I leave it open here exactly what the structural correlates of this distinction are – the minimal assumption would be that the relevant local domain is XP.

³ As more detailed research on tense and aspect morphology has progressed, this view has become somewhat more articulated, with at least two levels of relation to tense being proposed (following on from and modifying original insights in Reichenbach 1947): (i) the establishment of a reference time; (ii) the anchoring of that reference time to the speech time. According to Giorgi and Pianesi (1997); Klein (1994); and Demirdache and Uribe-Etxebarria (2000), the intermediate reference time level is a syntactic head between the verbal domain and T that they call Asp(ect). The details of this more articulated view do not concern us here, as they concern the intricacies of manipulating the tense variable, which I will assume is built up from the eventive projection itself by a functional head that has the semantic effects of Krifka's temporal trace function.

⁴ Lexicalist approaches to the alternation tend to favor a subtractive approach whereby the external argument of a transitive verb is suppressed to give the inchoative alternant under certain conditions. Syntactic approaches lend themselves better to an 'additive' or structure building system. I will abstract away from this issue here, but see Ramchand (2007) for a discussion.

⁵ It is also important to distinguish these internal relationships from the event composition commonly represented in lattice-theoretic terms (Link 1983; Krifka 1989). Mereological composition within a lattice requires a symmetrical kind of 'join' operation, where every event, including internally complex ones, can in principle be 'joined' to any other. The event composition rule proposed here is asymmetrical, and the *init*, *proc*, *res* combination is maximal. Under this proposal, the intuition behind the event building portion of the clause cannot be expressed in lattice theoretic terms. Lattice theoretic approaches to events are relevant of course when applied to macroevents as atoms, and can be used to represent iterativity and plurality.

⁶ The discussion in this section also does not include a treatment of stative verbs of various kinds, whether they need to be distinguished into different varieties depending on the participant role of the subject, or indeed the proper treatment of the thematic role of experiencer. Stative verbs would possess more impoverished syntactic structure, lacking in particular the process projection under this system. Thus, the particular decompositions proposed here have unfortunately nothing to say about these other distinctions. It remains to be seen whether the different types of stative verb should be distinguished in an event syntactic way that goes beyond the very spare classification of verbs proposed here into states vs. dynamic events of particular types. See Belletti and Rizzi (1988) for a seminal article concerning the different argument structures of different kinds of psych verbs in Italian, and also Van Voorst (1992) and Pesetsky (1995) for important work in this area. Van Voorst (1992) interestingly argues that the existence of an experiencer role is less important for determining a verb's linguistic behavior than its basic aktionsart properties. Finally, see also Cinque (1990) for an argument that even adjectival predicates need to be distinguished according to whether their subject argument is more agent like or patient like (cf. the unergative vs. unaccusative distinction of Perlmutter 1978). A proper treatment of this question is beyond the scope of this article.

Works Cited

Asher, Nicholas. 1998. Events, facts, propositions and evolutive anaphora. Speaking of events, ed. by James Higginbotham, Fabio Pianesi and Achille Varzi. Oxford, UK: Oxford University Press.

- Baker, Mark. 1997. Thematic roles and syntactic structure. *Elements of grammar: handbook in generative syntax*, ed. by Liliane Haegeman, 73–137, Dordrecht, The Netherlands: Kluwer.
- Baker, Mark C. 2003. Verbs, nouns, and adjectives: their Universal Grammar, Number 102 in *Cambridge Studies in Linguistics*, Cambridge, UK: Cambridge University Press.
- Belletti, Adriana, and Luigi Rizzi. 1988. Psych-verbs and θ -theory. *Natural Language and Linguistic Theory* 6.291–352.
- Bennett, Jonathan. 1988. *Events and their names*. Oxford, UK: Clarendon Press.
- Borer, Hagit. 2005. *Structuring sense: an exo-skeletal trilogy*. New York, NY: Oxford University Press.
- Carlson, Gregory. 1998. Thematic roles and the individuation of events. *Events and grammar*, ed. by Susan Rothstein. Dordrecht, The Netherlands: Kluwer.
- Carlson, Gregory Norman. 1977. *Reference to kinds in English*. PhD thesis, University of Massachusetts at Amherst.
- Chierchia, Gennaro. 1998. Reference to kinds across languages. *Natural Language Semantics* 6.339–405.
- Cinque, Guglielmo. 1990. Ergative adjectives and the lexicalist hypothesis. *Natural Language and Linguistic Theory* 8.1–39.
- Davidson, Donald. 1967. The logical form of action sentences. *The logic of decision and action*, ed. by Nicholas Rescher. Pittsburgh, PA: University of Pittsburgh Press.
- Demirdache, Hamida, and Myriam Uribe-Etxebarria. 2000. The primitives of temporal relations. *Step by step. Essays on minimalist syntax in honour of Howard Lasnik*, ed. by Roger Martin, David Michaels and Juan Uriagereka, 157–86. Cambridge, MA: MIT Press.
- Dowty, David R. 1979. Word meaning and Montague grammar: the semantics of verbs and times in generative semantics and in Montague's PTQ. Dordrecht, The Netherlands: D. Reidel.
- . 1989. On the semantic content of the notion 'thematic role'. *Properties, types and meaning*, ed. by B. Partee G. Chierchia and R. Turner, 69–129, Dordrecht, The Netherlands: D. Reidel.
- Ernst, Thomas. 2002. *The syntax of adjuncts*. Cambridge, UK: Cambridge University Press.
- Filip, Hana. 1999. *Aspect, eventuality types, and nominal reference*. New York, NY: Garland.
- Giorgi, Alessandra, and Fabio Pianesi. 1997. *Tense and aspect: from semantics to morphosyntax*. New York, NY: Oxford University Press.
- Grimshaw, Jane. 1990. *Argument structure*. Cambridge, MA: MIT Press.
- Hacker, P. M. S. 1982. Events, ontology, and grammar. *Philosophy* 57.477–96.
- Hale, Ken, and Samuel J. Keyser. 1987. A view from the middle. *Lexicon Project Working Papers* 10. Cambridge, MA: Center for Cognitive Science, MIT.
- . 1993. On argument structure and the lexical expression of syntactic relations. *The view from Building 20: Essays in linguistics in honor of Sylvain Bromberger*, ed. by Kenneth Hale and Samuel Jay Keyser, 53–109, Cambridge, MA: MIT Press.
- Harley, Heidi. 1995. *Subjects, events, and licensing*. PhD thesis. Cambridge, MA: MIT.
- Higginbotham, James. 1985. On semantics. *Linguistic Inquiry* 16.547–93.
- . 2000. On events in linguistic semantics. *Speaking of events*, ed. by Fabio Pianesi and Achille Varzi, 49–79, New York, NY: Oxford University Press.
- . 2001. *Accomplishments*, ms., Oxford University.
- . 2005. Event positions: Suppression and emergence. *Theoretical Linguistics* 31.349–58.
- Higginbotham, James, and Gillian Ramchand. 1997. The stage-level/individual-level distinction and the mapping hypothesis. *Oxford Working Papers in Linguistics, Philology and Phonetics* 2.53–83.
- Hoekstra, Teun. 1988. Small clause results. *Lingua* 74.101–39.
- Hoekstra, Teun, and René Mulder. 1990. Unergatives as copular verbs. *Linguistic Review* 7.1–79.
- Jackendoff, Ray. 1990. *Semantic structures*. Cambridge, MA: MIT Press.
- Kamp, Hans, and Uwe Reyle. 1993. *From discourse to logic*. Dordrecht, The Netherlands: D. Reidel.
- Kim, Jaegwon. 1966. On the psycho-physical identity theory. *American Philosophical Quarterly* 3.277–85.
- . 1976. Events as property exemplifications. *Action theory*, ed. by M. Brand and D. Walton, Dordrecht, The Netherlands: D. Reidel.

- Klein, Wolfgang. 1994. *Time in language*. London: Routledge.
- Kratzer, Angelika. 1995. Stage-level and individual-level predicates. *The Generic Book*, ed. by Gregory Carlson and Francis Jeffrey Pelletier, 125–75, Chicago, IL: The University of Chicago Press.
- . 1996. Severing the external argument from the verb. *Phrase structure and the lexicon*, ed. by Johann Rooryck and Laurie Zaring, 109–37, Dordrecht, The Netherlands: Kluwer.
- Krifka, Manfred. 1989. Nominal reference and temporal constitution and quantification in event semantics. *Semantics and contextual expression*, ed. by P. van Emde, Boas R. Bartsch, J. van Bentham, 75–115, Dordrecht, The Netherlands: Foris.
- . 1992. Thematic relations and links between nominal reference and temporal constitution. *Lexical matters*, Ivan A. Sag and Anna Szabolcsi, 29–53, Stanford, CA: CSLI.
- Landman, Fred. 2000. *Events and plurality: the Jerusalem lectures*. Dordrecht, The Netherlands: Springer.
- Larson, Richard K. 1988. On the double object construction. *Linguistic Inquiry* 19.335–91.
- Lasnik, Peter. 1995. *Plurality, conjunction and events*. Dordrecht, The Netherlands: Kluwer.
- Levin, Beth, and Malka Rappaport Hovav. 1995. *Unaccusativity: at the syntax-lexical semantics interface*. Cambridge, MA: MIT Press.
- Link, G. 1983. The logical analysis of plurals and mass terms: A lattice-theoretical approach. *Meaning, use and interpretation of language*, ed. by R. Bäuerle, C. Schwarze, and A. von Stechow, 302–23, Berlin, Germany: Walter de Gruyter.
- Maienborn, Claudia. 2005. On the limits of the davidsonian approach. *Theoretical Linguistics* 31.275–317.
- Marantz, Alec. 1984. *On the nature of grammatical relations*. Cambridge, MA: MIT Press.
- Parsons, Terence. 1990. *Events in the semantics of English: a study in subatomic semantics*. Cambridge, MA: MIT Press.
- Perlmutter, David. 1978. Impersonal passives and the unaccusative hypothesis. *Proceedings of the Fourth Annual Meeting of the Berkeley Linguistics Society* 4.157–89.
- Pesetsky, David. 1995. *Zero syntax: experiencers and cascades*. Cambridge, MA: MIT Press.
- Pollock, Jean Yves. 1989. Verb movement, universal grammar, and the structure of IP. *Linguistic Inquiry* 20.365–424.
- Pustejovsky, James. 1991. The syntax of event structure. *Cognition* 41.47–81.
- Pylkkänen, Liina. 1999. Causation and external arguments. *Papers from the UPenn/MIT Roundtable on the lexicon*, ed. by Liina Pylkkänen, Angeliek van Hout and Heidi Harley, 161–83, Cambridge, MA: MITWPL 35.
- Quine, W. V. 1960. *Word and object*. Cambridge, MA: MIT Press.
- Ramchand, Gillian. 1996. Two subject positions in Scottish Gaelic. *Natural Language Semantics* 4.165–191.
- . 2007. *Verb meaning and the lexicon*. Forthcoming from Cambridge University Press, available from <http://ling.auf.net/lingBuzz/000307>.
- Rappaport Hovav, Malka, and Beth Levin. 1998. Building verb meanings. *The projection of arguments: lexical and compositional factors*, ed. by Miriam Butt and Wilhelm Geuder, 97–134, Stanford, CA: CSLI publications.
- Reichenbach, Hans. 1947. *Elements of symbolic logic*. New York, NY: Macmillan.
- Reinhart, T. 2002. The theta system – an overview. *Theoretical Linguistics* 28.229–90.
- Ritter, Elizabeth, and Sara Thomas Rosen. 1998. Delimiting events in syntax. *The projection of arguments: lexical and compositional factors*, ed. by Miriam Butt and Wilhelm Geuder, 135–64, Stanford, CA: CSLI Publications.
- Rizzi, Luigi. 1997. The fine structure of the left periphery. *Elements of grammar: handbook in generative syntax*, ed. by Liliane Haegeman, 281–337, Dordrecht, The Netherlands: Kluwer.
- Schein, Barry. 1993. *Plurals and events*. Cambridge, MA: MIT Press.
- . 2002. Events and the semantic content of thematic relations. *Logical form and language*, ed. by Gerhard Preyer and Georg Peter, 263–344, Oxford, UK: Oxford University Press.
- de Swart, Henriette. 1996. (In)definites and genericity. *Quantifiers, deduction and context*, ed. by C. Pinñon, M. Kanazawa and H. de Swart, 171–94, Stanford, CA: CSLI Publications.
- . 1998. Aspect shift and coercion. *Natural Language and Linguistic Theory* 16.347–85.

- Szabolcsi, Anna. 1994. The noun phrase. *Syntax and semantics* 27. The syntactic structure of Hungarian, ed. by F. Kiefer and K. É. Kiss, 179–274, San Diego, CA: Academic Press.
- Travis, Lisa. 2000. Event structure in syntax. Events as grammatical objects: the converging perspectives of lexical semantics and syntax, ed. by Carol Tenny and James Pustejovsky, 145–185. Stanford, CA: CSLI.
- Van Voorst, Jan. 1992. The aspectual semantics of psychological verbs. *Linguistics and Philosophy* 15.65–92.
- Vendler, Zeno. 1967. *Linguistics in Philosophy*. Ithaca, NY: Cornell University Press.
- Verkuyl, Henk J. 1972. On the compositional nature of the aspects. Dordrecht, The Netherlands: D. Reidel.
- . 1993. *A theory of aspectuality: the interaction between temporal and atemporal structure*. Cambridge, UK: Cambridge University Press.
- Williams, Edwin. 1982. The NP cycle. *Linguistic Inquiry* 13.277–95.